AP Precalculus — In-Class Test

Rational Functions & Long Division (30 problems)

Name:

Quick Review (no calculus)

- **Polynomial long division:** $\dfrac{N(x)}{D(x)} = Q(x) + \dfrac{R(x)}{D(x)}$ where $\deg R < \deg D$. This gives an equivalent representation.
- **Zeros (x-intercepts):** Solve $N(x) = 0$ *after* canceling any common factors with $D(x)$. Canceled factors $\Rightarrow$ no intercept at that x (usually a hole).
- **Vertical asymptotes (VA):** Values of x that make $D(x) = 0$ *after* canceling common factors.
- **Holes (removable):** Values of x where a common factor cancels; point is missing but no VA.
- **End behavior (as $x \to \pm\infty$):**
  - $\deg N < \deg D \Rightarrow y \to 0$ (HA: $y = 0$).
  - $\deg N = \deg D \Rightarrow$ HA $y =$ (leading coeff of N) / (leading coeff of D).
  - $\deg N = \deg D + 1 \Rightarrow$ slant (oblique) asymptote = quotient of long division.
  - $\deg N > \deg D + 1 \Rightarrow$ end behavior follows the polynomial part from division.

1. (3 points) Divide and write in the form $Q(x) + \dfrac{R(x)}{D(x)}$: $\dfrac{2x^2 + 5x + 3}{x + 2}$.

Show your work here

2. (3 points) Long divide: $\dfrac{3x^3 - 7x + 10}{x - 1}$.

Show your work here

3. (3 points) Long divide: $\dfrac{x^3 - 4x^2 + 6x - 5}{x - 2}$.

Show your work here

4. (3 points) Use division to rewrite $\frac{2x^2 - 9}{x - 3}$ and state the end behavior of the rational function.

Show your work here

5. (3 points) Perform division: $\frac{4x^3 + 8x^2 - 5}{2x + 1}$.

Show your work here

6. (3 points) Write $\frac{x^2 + 1}{x}$ as polynomial + proper fraction and give its end behavior.

Show your work here

7. (3 points) Find the x-intercepts of $f(x) = \frac{(x-3)(x+1)}{x-2}$.

Show your work here

8. (3 points) Find the x-intercepts of $g(x) = \dfrac{(x-4)(x+2)}{(x-4)(x-1)}$.

Show your work here

9. (3 points) Determine all zeros of $h(x) = \dfrac{x(x-5)^2}{(x+3)}$ and their multiplicities as intercepts.

Show your work here

10. (3 points) True/False (justify): If $N(x)$ and $D(x)$ share a factor $(x-a)$, then $x=a$ *cannot* be an x-intercept of the rational function.

Show your work here

11. (3 points) Find the x-intercepts of $r(x) = \dfrac{x^2-9}{x^2-1}$ after simplifying.

Show your work here

12. (3 points) Find all x-intercepts of $p(x) = \dfrac{(x+2)^2(x-1)}{(x+2)(x+5)}$.

Show your work here

13. (3 points) State the vertical asymptotes of $f(x) = \dfrac{1}{x^2-4}$.

Show your work here

14. (3 points) Vertical asymptotes of $g(x) = \dfrac{(x-3)}{(x-3)(x+1)}$.

Show your work here

15. (3 points) Vertical asymptotes of $h(x) = \dfrac{(x+2)}{(x+2)^2(x-5)}$.

Show your work here

16. (3 points) Does $r(x) = \dfrac{(x-1)(x+4)}{(x-1)}$ have any vertical asymptotes? Explain.

Show your work here

17. (3 points) Find all vertical asymptotes of $q(x) = \dfrac{x^2-1}{x(x-1)}$ (simplify first if needed).

Show your work here

18. (3 points) Determine the vertical asymptotes of $s(x) = \dfrac{(x-2)^2}{(x-2)(x+3)}$.

Show your work here

19. (3 points) Find the location (x-value) of any hole in $f(x) = \dfrac{(x+1)(x-2)}{(x-2)(x-3)}$. Does a VA remain at that x?

Show your work here

20. (3 points) Identify holes (if any) for $g(x) = \dfrac{(x-5)^2}{(x-5)(x+1)}$, and state the simplified function.

Show your work here

21. (3 points) Determine all holes of $h(x) = \dfrac{x^2-4}{x^2-5x+6}$.

Show your work here

22. (3 points) Does $r(x) = \dfrac{(x-3)}{(x-3)^2}$ have a hole at $x = 3$? Explain.

Show your work here

23. (3 points) End behavior (HA or slant?) for $f(x) = \dfrac{3x^2+1}{x^2-4}$.

Show your work here

24. (3 points) End behavior for $g(x) = \dfrac{2x^3 - 5}{x^2 + 1}$ (name any slant or describe growth).

Show your work here

25. (3 points) End behavior for $h(x) = \dfrac{4x - 7}{2x^2 + 3}$.

Show your work here

26. (3 points) End behavior for $r(x) = \dfrac{x^4 - 9}{x^2 - 1}$ (use long division idea).

Show your work here

27. (3 points) End behavior for $q(x) = \dfrac{x^2 + 5x + 6}{x - 1}$ (state slant if any).

Show your work here

28. (4 points) For $F(x) = \dfrac{(x-2)(x+1)}{(x-2)(x-4)}$: list holes, vertical asymptotes, and x-intercepts (after simplifying).

Show your work here

29. (4 points) A rational function has a hole at $x = 1$, a vertical asymptote at $x = -2$, and x-intercepts at $x = 3$ and $x = -4$. Give one possible factored form $\dfrac{N(x)}{D(x)}$ (lowest degrees you can).

Show your work here

30. (4 points) Sketch-logic (no graphing): For $S(x) = \dfrac{(x+3)^2(x-1)}{(x-2)(x+1)^2}$, state: HA/slant, VAs, holes, x-intercepts with bounce/cross, and whether $y \to 0$, constant, or polynomial at infinity.

Show your work here